

5	(a) diffraction: <u>spreading/diverging</u> of <u>waves/light</u> (takes place) at (each) slit/ element/gap/aperture	B1	
	interference: overlapping of waves (from coherent sources at each element)	B1	
	path difference λ /phase difference of $360(^{\circ})/2\pi$ (produces the first order)	B1	[3]
	(b) $d \sin \theta = n\lambda$ or $\sin \theta = Nn\lambda$	C1	
	$d = (2 \times 486 \times 10^{-9}) / \sin 29.7^{\circ} (= 1.962 \times 10^{-6})$	C1	
	number of lines = 510 (509.7) mm^{-1}	A1	[3]

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2016	9702	22